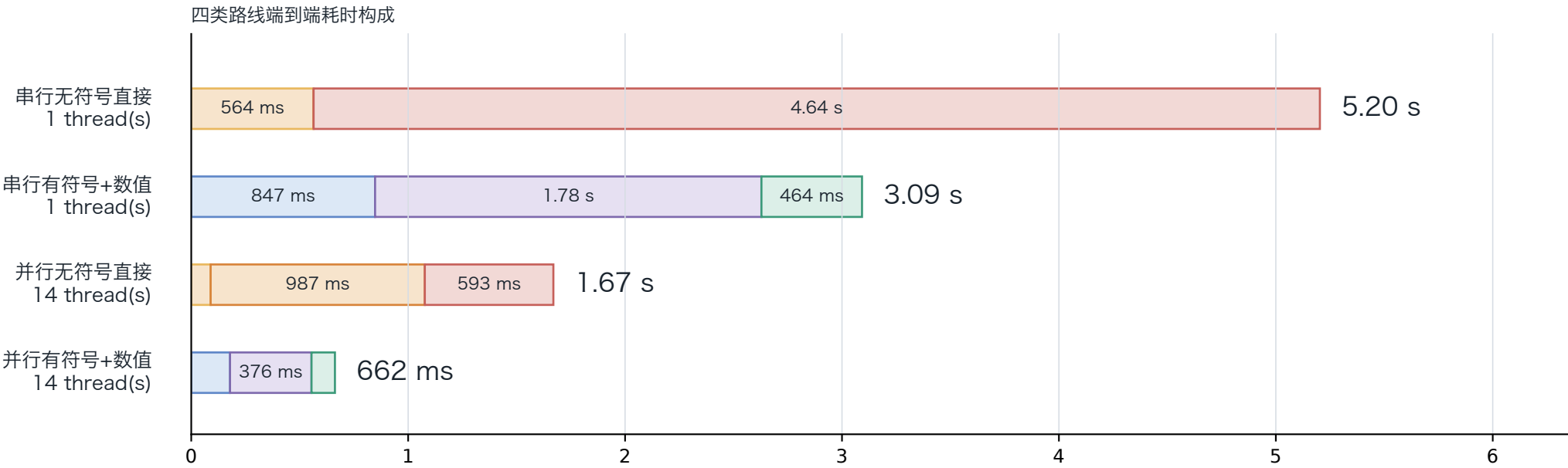
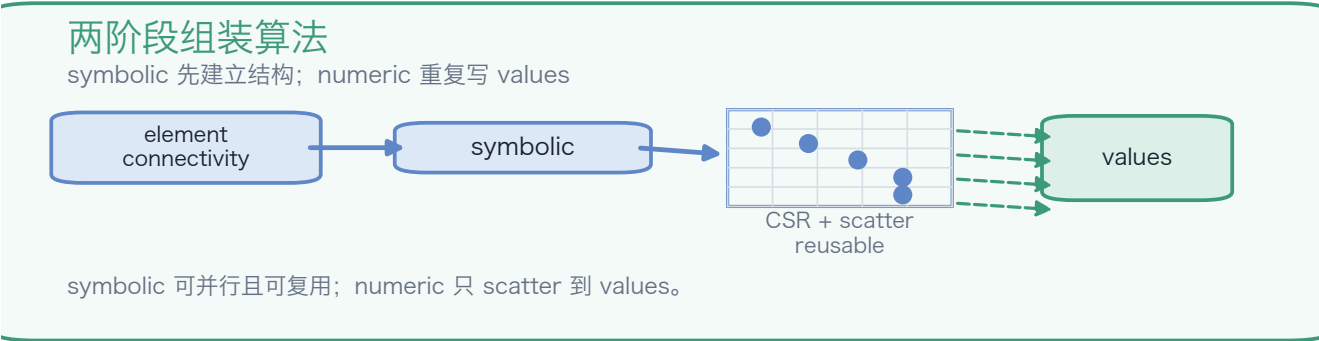
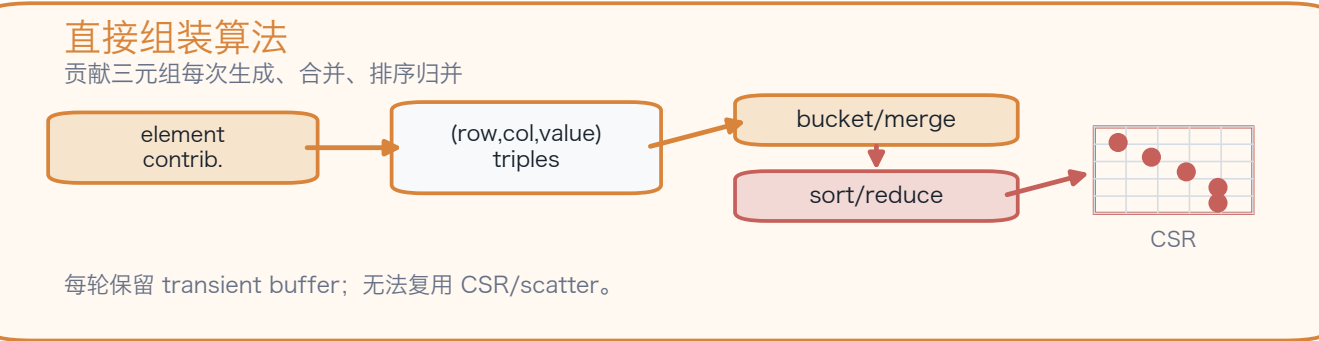


四类刚度组装路线：时间优先，内存为辅

WindHub Tet4 / Apple M4 Max; total time includes symbolic/direct construction and numeric assembly



对比结论

1.68x

串行：有符号优于无符号
5.20 s → 3.09 s

4.67x

并行符号优于串行符号
3.09 s → 662 ms

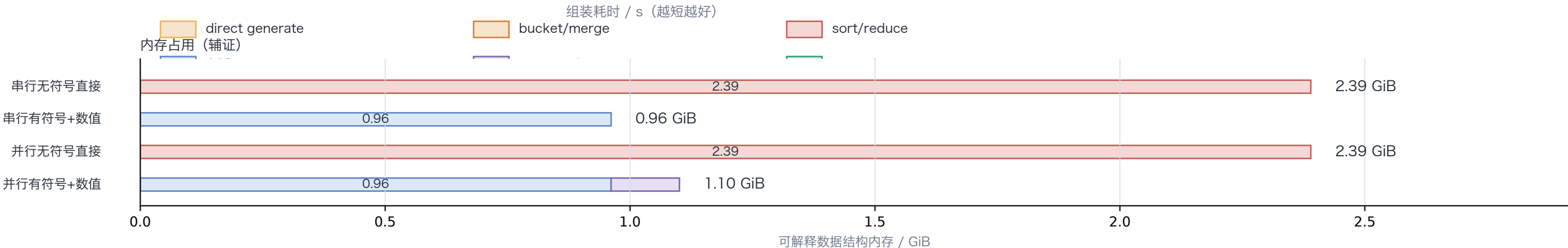
2.52x

同为 14 线程：有符号优于 direct
1.67 s → 662 ms

7.85x

最佳路线相对串行 direct
5.20 s → 662 ms

最佳路线 rel_L2 = 1.5e-16



Source: curated WindHub / Apple M4 Max quadrant rows; direct/no-symbolic is contribution-list sort/reduce, not a dense matrix.

注：不是 OS RSS; direct 主要是 transient contribution buffer. symbolic 主要是可复用 CSR/scatter。